

A simple and inherently privative meaning for *fake**

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1. Introduction

Adjectives like *fake*, *counterfeit*, *artificial* are commonly said to fall under the class of privatives. Privative adjectives are usually defined via their characteristic entailment pattern:

$$(1) \quad \llbracket Adj\ N \rrbracket \not\subset \llbracket N \rrbracket.$$

Other adjectival classes include the following: intersectives, whose output sits at the intersection between the adjective and the noun (e.g. *French*); subsectives, which don't denote an independent set but restrict noun extensions to a subset (e.g. *good*); and plain non-subsectives, whose output could be inside or outside the extension of the input noun (e.g. *alleged*) (cf. Partee, 2010, a.o.). In a way, the challenge posed specifically by privative modification consists in explaining how fake *Ns* are related to *Ns* without *being Ns*. In this paper, I will review what I take to be the two main approaches to this challenge, lay out their shortcomings, and briefly sketch a novel proposal.

Partee (2010) proposed that *fake* is actually not privative, but subsective, like *good*. Subsective adjectives are defined by the entailment pattern below:¹

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¹One may view these, as Partee suggests, as meaning postulates on different subclasses of adjectives.

$$(2) \quad \llbracket \text{Adj } N \rrbracket \subseteq \llbracket N \rrbracket.$$

Fake is a subjective ‘in disguise’ in that it operates on *broadened* extensions. Because there are no fake guns in the extension of guns literally intended, *fake* operates on the extension of gun-like things, thanks to specific pragmatic principles. It then restricts this broadened set to the subset that only encompasses fake guns. This allows us to keep (i) a very simple lexical entry for *fake* and (ii) a neatly constrained classification of modifiers.

Others have chosen the exact opposite path: Del Pinal (2015) proposes that *fake* is truly privative in that it takes an extension as an input and returns an extension that is not included in the input extension. Moreover, he argues that the semantic complexity of *fake* forces us to abandon the traditional notion of purely extensional NP meanings. His Dual Content Semantics (DCS) (Del Pinal, 2015; 2018) has it that NPs have a bi-dimensional structure: the E-structure of nouns determines extensions, while the C-structure incorporates the related “core facts”, a set of beliefs about the extension organised around different dimensions.

In this paper, I propose an intermediate view: contra Del Pinal and with Partee, I propose there is no need to move to richer lexical meanings to account for *fake*: a one-dimensional semantics suffices. Contra Partee and with Del Pinal, I propose the entailment pattern characteristic of *fake* derives from its meaning, rather than from its interaction with pragmatic principles of interpretation. In a nutshell, I propose that ‘fake’ has the simple meaning paraphrased in (3) below:

$$(3) \quad \llbracket \text{fake gun} \rrbracket = \lambda x. x \text{ is intended to look/seem like a gun and } x \text{ is not a gun}$$

One thing that teases apart the predictions of the present approach from Partee’s and Del Pinal’s theories are its predictions for cases of multiple application of *fake*. Iterated application is an excellent testbed for theories of privative modification, as it makes entailment patterns unstable (cf. Jespersen *et al.*, 2017). While the modifier *non-* and *fake* both give rise to the same privative entailment pattern when applied once, they differ when applied twice.

$$(4) \quad x \text{ is a non-gun} \models x \text{ is not a gun} \qquad (5) \quad x \text{ is a fake gun} \models x \text{ is not a gun}$$

$$(6) \quad x \text{ is a non-non-gun} \models x \text{ is a gun} \qquad (7) \quad x \text{ is a fake fake gun} \not\models x \text{ is a gun}$$

If you are unsure about (7), notice that (8a) qualifies as a gun, but not (8b). Yet both are fake fake guns.

- (8) a. FAKE FAKE GUN; GUN.
An airsoft gun made to shoot actual bullets and kill in an airsoft game.
b. FAKE FAKE GUN; NON-GUN.
A cake that was built to resemble a famous model of a toy gun.

In what follows, I show why neither Partee’s nor Del Pinal’s account provide adequate predictions for these cases, and then sketch my proposal.

2. Partee 2010

Partee (2010) does not make fine-grained enough predictions to account for iteration. She proposes that two pragmatic principles yield the interpretation of privatives.

(9) **Head Primacy (HP):**

An *Adj N* is a *N*.

Or: in a modifier-head structure, the modifier is the modifier and the head the mod-
ifiee.

(10) **Non-Vacuity (NV):**

A structure *Adj N* should have a non-empty positive extension and a non-empty
negative extension.

In privative modification, because the positive extension of *Adj N* is empty, Head Primacy is given up in order not to violate Non-Vacuity: the extension of *N* is stretched to include ‘*N*-like things’ of which *Adj* is true. A clear illustration of this reasoning can be given for expressions like *stone lion*:

(11) ‘Stone lion’

Try to interpret the expression as $\{\text{stone lion}\} \subseteq \{\text{lion}\}$, get an empty
extension

⇒ Non-Vacuity fails.

Give up Head Primacy: a stone lion is now a stone.

The extension of ‘lion’ is broadened to ‘lion-like things’ that include
things made of stone.

Partee argues that Head Primacy and Non-Vacuity, *qua* constraints on the possible meanings of *Adj-N*, could be seen as presuppositions that must be satisfied by any use of an adjective. In that case, coercions such as the one happening in (11) could be seen as a form of presupposition accommodation.

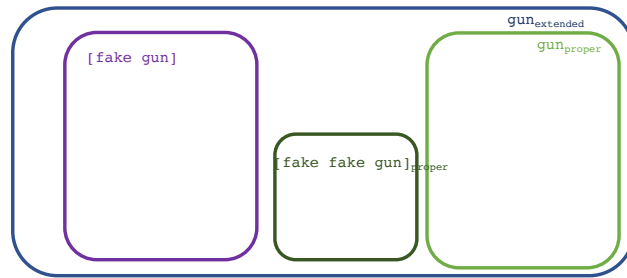
This allows us to well-predict the behavior of *stone lion*, as in (11). There are however reasons to think that this is not sufficient to explain the behavior of *fake*. Indeed, while it seems obvious why the intersection between *stone* and *lion* is (at first) vacuous – viz., there are no biological lions made of stone – it is far less obvious why *fake* would yield a vacuous extension when applied to *gun*. In other words, we need a more precise lexical entry for *fake* to understand why this pragmatic reasoning is triggered in the first place.² In a way, then, while HP and NV seem to be real principles, Partee’s account is not specified enough.

² Additionally, notice that *stone* only triggers this reasoning with certain *N*s, while with others composition can proceed smoothly: a stone statue is still a statue. *Fake*, instead, does so with any *N* it takes as an input. In this framework, then, this would mean that the grammar contains a word that systematically violates Non-Vacuity, i.e. that systematically yields presupposition failures.

This emerges clearly when we look at iteration. To see this, let us go through the reasoning for the case of *fake fake gun*.

First, we compose *fake gun* in a way akin to what we did for *stone lion*: because *fake gun* would yield an empty extension if *gun* were interpreted literally, we broaden the extension of *guns* to include both real and fake guns. We then restrict this to only fake guns. Now we apply *fake* for a second time, and we get an empty extension again:

- (12) There are no fake [fake guns] in the extension of *fake gun*.



A similar reasoning is then triggered again:

- (13) Relax Head Primacy. $\llbracket \text{fake gun} \rrbracket$ is broadened to include both real and fake [fake guns] (just like in the computation of ‘fake gun’ the denotation of ‘gun’ was extended to include both real and fake guns).

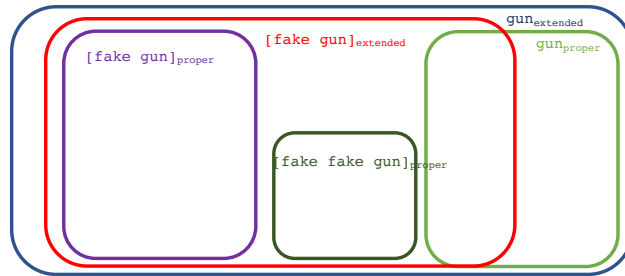


Figure 1: A representation of the step in (13) relax head primacy and broaden the extension of fake-guns-proper to fake-guns-extended, to include both real and fake [fake guns].

We know from the introduction that a fake fake gun may or may not be a gun (but see section 4 for a more detailed discussion of the data). This prediction is not made by Partee’s account, because we don’t know precisely how the denotation of ‘fake gun’ is broadened to include fake fake guns. That is, we could be in either of the three scenarios in Figures 2, 3 or 4 below.

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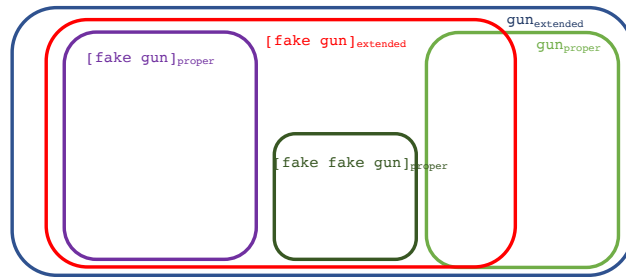


Figure 2: *A scenario in which all fake fake guns are non-guns.*

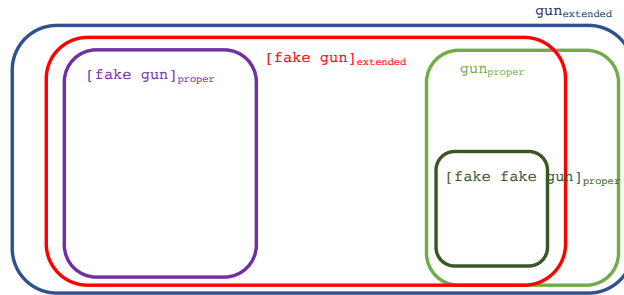


Figure 3: *A scenario in which all fake fake guns are guns-proper.*

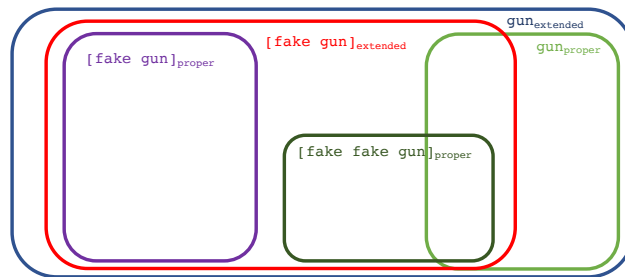


Figure 4: *A scenario in which a fake fake gun may or may not be a gun-proper.*

3. Del Pinal 2015

In Dual Content Semantics, meaning is determined along two lines: the E-structure of nouns determines their extension, while the C-structure incorporates the related ‘core facts’,

a set of beliefs about the extension, in the form of *qualia*. This allows DCS to adress two opposing constraints: on the one hand, it seeks to solve puzzles related to lexical flexibilities. For instance, it can predict quite precisely the possible senses of the expression *stone lion*. On the other hand, it does not overgenerate, a problem that other theories arguing for a richer notion of lexical meaning have faced (see e.g. Asher 2011 for a review of such problems for contextualist theories of meaning), precisely because meaning is determined along fixed dimensions via specific compositional mechanisms. The meaning of *gun*, for instance, is as follows:

$$(14) \quad \llbracket \text{gun} \rrbracket_M^c =$$

E-structure:

$$\lambda x. \text{GUN}(x)$$

C-structure:

$$\text{C: } \lambda x. \text{PARTS_GUN}(x)$$

$$\text{P: } \lambda x. \text{PERCEPTUAL_GUN}(x)$$

$$\text{T: } \lambda x. \text{GEN } e [\text{SHOOTING}(e) \wedge \text{INSTRUMENT}(e, x)]$$

$$\text{A: } \lambda x. \exists e_1 [\text{MAKING}(e_1) \wedge \text{GOAL}(e_1, \text{GEN } e (\text{SHOOTING}(e) \wedge \text{INSTRUMENT}(e, x)))]$$

Del Pinal posits that ‘fake’ negates the E-structure and ‘promotes’ content from the C-structure to the E-structure: E-structurally, a fake gun isn’t a gun (negation of E-structure of ‘gun’), does not have the origin of a gun, has the function of having the shape of a gun, and does not have the purpose of a gun (C-structure of ‘gun’ manipulated and uploaded to E-structure of ‘fake gun’).

$$(15) \quad \llbracket \text{fake gun} \rrbracket_M^c =$$

E-structure:

$$\lambda x. \neg E(\llbracket \text{gun} \rrbracket_M^c)(x) \wedge$$

$$\neg Q_A(\llbracket \text{gun} \rrbracket_M^c)(x) \wedge$$

$$\exists e_2 [\text{MAKING}(e_2) \wedge \text{GOAL}(e_2, Q_P(\llbracket \text{gun} \rrbracket_M^c)(x))]$$

C-structure:

$$\text{C: } \lambda x. Q_C(\llbracket \text{gun} \rrbracket_M^c)(x)$$

$$\text{P: } \lambda x. Q_P(\llbracket \text{gun} \rrbracket_M^c)(x)$$

$$\text{T: } \lambda x. \lambda x. \neg Q_T(\llbracket \text{gun} \rrbracket_M^c)(x) \wedge Q_P(\llbracket \text{gun} \rrbracket_M^c)(x)$$

$$\text{A: } \lambda x. \exists e_2 [\text{MAKING}(e_2) \wedge \text{GOAL}(e_2, Q_P(\llbracket \text{gun} \rrbracket_M^c)(x))]$$

Martin (2018) showed that Del Pinal’s account, formulated as is, gives rise to a contradiction when ‘fake’ is applied twice (see the Appendix for a detailed explanation of why this happens).

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The source of the problem seems to be that DCS has ‘fake’ pass through the C-structure of its input NP: fake guns have mostly the same associated properties as guns. Besides not working compositionally in iterated application, this doesn’t seem to capture the (enriched) meaning of *fake* correctly. Fake guns don’t only have some properties of guns. They also have properties of their own: for instance, being made of plastic is more typical of a fake gun than of a gun. But how are we to determine these properties explicitly? Nothing in the meaning of *fake* and *gun* tells us that fake guns are commonly made of plastic. Their composition simply tells us that fake guns will be made of a material that resembles the material of which guns are made, *whatever these materials may be*. It is a merely contingent fact that many gun replicas are made of plastic, and not of another less common metal-resembling material. Then, to know what material typically makes up fake guns, we cannot but look at *actual* instances of fake guns. What is tricky is that precisely these ‘emergent’ properties may be needed for further compositional steps. For instance, a *fake fake* gun may simulate exactly the property of being made of plastic: it may be fakely made of plastic, thus for instance of plastic-resembling metal. This suggests that the C-structure is not compositionally determined, and has to be newly computed at each step in a composition. Martin provides a local fix to DCS for this problem, but given the above considerations, it is licit to ask if we truly need a C-structure to account for ‘fake’.

4. Proposal

Similarity is an excellent tool available to get access to ‘diagnostic’ properties. I propose:

$$(16) \quad \llbracket \text{fake} \rrbracket_w = \lambda P. \lambda x. \text{INTENDED}_w(x) (\lambda y. \text{SEEM-LIKE}_w(y, P)) \wedge \neg P_w(x)$$

INTENDED here stands for a function that for any input pair of an entity x and a predicate P , will output the proposition that there was an action intentionally performed on x whose goal was that x seem like a P . One way to implement this would as follows:

$$(17) \quad \begin{array}{ll} \text{a.} & \text{INTENDED}(x)(P) = \exists e. \text{ACTION}(e, x) \wedge \text{GOAL}(e, P(x)) \\ \text{b.} & \text{INTENDED}(x) (\lambda y. \text{SEEM-LIKE}(y, \text{gun})) = \\ & \exists e. \text{ACTION}(e, x) \wedge \text{GOAL}(e, \text{SEEM-LIKE}(x, \text{gun})) \end{array}$$

In words, there is an event that is an action performed on x whose goal is that x seem like a P . This action needs to be intentional, as not any gun-resembling thing qualifies as a fake gun. For instance, a configuration of atoms on Mars that happens to look exactly like a gun without being one doesn’t qualify as a fake gun. I leave GOAL as a black box, but a semantics in terms of bouletic modality should be straightforward.

At this point, if we reduce the semantics of ‘fake’ partly to the semantics of similarity verbs like ‘resemble’ or ‘seem like’, we can import the mechanisms by which similarity verbs get access to the diagnostic properties of a category. And if similarity verbs can be accounted for by a one-dimensional semantics, as they recently have been (cf. Guerrini, 2022) we can dispense completely with the C-structure when accounting for ‘fake’ – but

of course, that Del Pinal’s DCS is not needed for ‘fake’ does not mean it is not needed *tout court*.

4.1 An illustration with Guerrini’s (2022) semantics for similarity constructions.

Putting aside the details of his compositional implementation, Guerrini proposes that the logical form for sentences like ‘John resembles a lawyer’ is as follows:

$$(18) \quad \text{GEN}[lawyer(x)] \left[\text{for all relevant partitions } \underline{A} \text{ on the universe of individuals, John falls in the same cell of } \underline{A} \text{ as } x \right]$$

The presence of GEN accounts for a number of facts, e.g. the intuitively generic interpretation of the indefinite: resembling a lawyer is approximately synonymous to resembling a *typical* lawyer. The rest of the (here informal) logical form is based on his account of simpler sentences like ‘John resembles Mary’:

$$(19) \quad \text{For all relevant partitions } \underline{A} \text{ on the universe of individuals, John falls in the same cell of } \underline{A} \text{ as Mary.}$$

I now abbreviate informal logical forms like (19) as ‘John SRP Mary’ (for ‘John shares relevant properties with Mary’). Now, replacing a logical form parallel to (18) for SEEM-LIKE in the lexical entry in (16), we get:

$$(20) \quad \llbracket \text{fake} \rrbracket_w = \lambda P. \lambda x. \text{INTENDED}(x) (\lambda z. \text{GEN}[P_w(y)] [z \text{ SRP } y]) \wedge \neg P_w(x)$$

We are thus able to make fine-grained predictions about the meaning of ‘fake’ without having to postulate an additional layer of meaning for lexical items. And incidentally, the presence of genericity seems to be justified in ‘fake’: someone cannot qualify as a fake lawyer in virtue of intending to resemble a specific, atypical lawyer; they have to intend to share a relevant set of properties with a *typical* lawyer.

4.2 Iteration in my proposal

But regardless of what substance one builds into the SEEM-LIKE black box, the semantics in (16) makes fine-grained, correct predictions about double (and iterated) application of ‘fake’. A fake fake gun is not a fake gun, and is intended to look like a fake gun, as in (21), which can be further developed into (22). Henceforth, to simplify the notation I will write $\text{INTENDED}(P(x))$, instead of $\text{INTENDED}(x) (\lambda y. P(y))$.

$$(21) \quad \llbracket \text{fake fake gun} \rrbracket = \begin{array}{ll} \text{i. } \lambda x. \neg \text{fake-gun}(x) \wedge \\ \text{ii. } \text{INTENDED}(\text{SEEM-LIKE}(x, \text{fake-gun})) \end{array}$$

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$$(22) \quad \llbracket \text{fake fake gun} \rrbracket = \begin{array}{l} \text{i. } \neg(\neg \text{gun}(x) \wedge \text{INTENDED}(\text{SEEM-LIKE}(x, \text{gun}))) \wedge \\ \text{ii. } \text{INTENDED}(\text{SEEM-LIKE}(x, \text{fake-gun})) \end{array}$$

This yields a complex truth table predicting that three classes of objects should qualify as fake fake guns.

	$\mathcal{G}_1$	$\mathcal{G}_2$	$\mathcal{G}_3$
$\neg \text{gun}(x)$	0	1	0
$\text{INTENDED}(\text{SEEM-LIKE}(x, \text{gun}))$	1	0	0
$\text{INTENDED}(\text{SEEM-LIKE}(x, \text{fake gun}))$	1	1	1

Let us check that this is right for all three classes.

Class $\mathcal{G}_1$

An object in $\mathcal{G}_1$ is a gun, intended to resemble a gun, and intended to resemble a fake gun. In other words: the object is intended to look like a gun precisely because *successful* fake guns look like guns; however, it turns out it actually is a gun, thus not a fake gun. For instance, take the object described in (8a). This object falls in this class, and also intuitively qualifies as a fake fake gun.

Class $\mathcal{G}_2$

An object in the $\mathcal{G}_2$ class, instead, is: not a gun; not intended to look like a gun; intended to look like a fake gun. An object that falls in this class is the cake described above in (8b), reported below in (23).

- (23) FAKE FAKE GUN; NON-GUN.
A cake that was built to resemble a famous model of a toy gun.

To give a more concrete example of $\mathcal{G}_2$ -type cases outside the realm of guns, consider the following context.

- (24) *Context:* The CIA has placed an agent inside the KGB, but the KGB found out about it and is on the hunt. John, an old CIA sleeper agent, decides to sacrifice himself to save the mole. He starts behaving like an American undercover would: he gives excessive detail on his Russian origins, and is very theatrical about having a Russian cultural background.

Now observe the contrast between (25a) and (25b) (and, incidentally, (25c), which shows that *fake* remains truly privative in iteration).

- (25) a. John is a fake fake KGB agent. TRUE

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- | | | |
|----|---------------------------|-------|
| b. | John is a KGB agent. | FALSE |
| c. | John is a fake KGB agent. | FALSE |

Class $\mathcal{G}_3$

Lastly, an object in the $\mathcal{G}_3$ class is: a gun; not intended to look like a gun; intended to look like a fake gun. Imagine terrorists who want to board a flight with a gun, and to this effect produce a real gun that looks like a toy gun. My account correctly predicts this object to qualify as a fake fake gun, as it is a gun, but not intended to look like one: instead, it is intended to look like something that qualifies as a fake gun, a toy gun.

5. Conclusion

I have argued for a one-dimensional, intrinsically privative semantics for ‘fake’. Basing ‘fake’ on similarity allows us to see that the considerable semantic complexity of this adjective is actually mostly compositionally inert. Precisely *what* is fake is very open-ended, because similarity is a contingent notion: a fake *X* may simulate properties that a *X* merely *happens* to have. More precisely, I argued that a fake *X* is (i) intended to look like a *X* and (ii) is not an *X*. I showed that this makes right predictions for iterated application of ‘fake’, as well as for the stacking of ‘fake’ on top of ‘typical’.

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Appendix

Del Pinal’s entry for *fake gun* reads as follows:

$$(26) \quad \llbracket \text{fake gun} \rrbracket_M^c =$$

E-structure:

$$\begin{aligned} & \lambda x. \neg E(\llbracket \text{gun} \rrbracket_M^c)(x) \wedge \\ & \neg Q_A(\llbracket \text{gun} \rrbracket_M^c)(x) \wedge \\ & \exists e_2 [\text{MAKING}(e_2) \wedge \text{GOAL}(e_2, Q_P(\llbracket \text{gun} \rrbracket_M^c)(x))] \end{aligned}$$

C-structure:

$$\begin{aligned} \text{C: } & \lambda x. Q_C(\llbracket \text{gun} \rrbracket_M^c)(x) \\ \text{P: } & \lambda x. Q_P(\llbracket \text{gun} \rrbracket_M^c)(x) \\ \text{T: } & \lambda x. \lambda x. \neg Q_T(\llbracket \text{gun} \rrbracket_M^c)(x) \wedge Q_P(\llbracket \text{gun} \rrbracket_M^c)(x) \\ \text{A: } & \lambda x. \exists e_2 [\text{MAKING}(e_2) \wedge \text{GOAL}(e_2, Q_P(\llbracket \text{gun} \rrbracket_M^c)(x))] \end{aligned}$$

Informally, the E-structure of ‘fake gun’ reads as follows. A fake gun:

- is not a gun
- does not have the origin of a gun
- does not have the purpose of a gun
- is an object x that has a building event that had as a goal that $\text{PERCEPTUAL_GUN}(x)$, i.e. that x look like an gun.

Martin (2018) observes that applying ‘fake’ to (26), we get a contradiction.

$$(27) \quad \llbracket \text{fake fake gun} \rrbracket_M^c =$$

E-structure:

$$\begin{aligned} & \lambda x. \neg E(\llbracket \text{fake gun} \rrbracket_M^c)(x) \wedge \\ & \neg Q_A(\llbracket \text{fake gun} \rrbracket_M^c)(x) \wedge \\ & \exists e_3 [\text{MAKING}(e_3) \wedge \text{GOAL}(e_3, Q_P(\llbracket \text{fake gun} \rrbracket_M^c)(x))] \end{aligned}$$

C-structure:

$$\begin{aligned} \text{C: } & \lambda x. Q_C(\llbracket \text{fake gun} \rrbracket_M^c)(x) \\ \text{P: } & \lambda x. Q_P(\llbracket \text{fake gun} \rrbracket_M^c)(x) \\ \text{T: } & \lambda x. \neg Q_T(\llbracket \text{fake gun} \rrbracket_M^c)(x) \\ \text{A: } & \lambda x. \exists e_3 [\text{MAKING}(e_3) \wedge \text{GOAL}(e_3, Q_P(\llbracket \text{fake gun} \rrbracket_M^c)(x))] \end{aligned}$$

After some calculations, we get something more legible:

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(28) $\llbracket \text{fake fake gun} \rrbracket_M^c =$

E-structure:

$$\begin{aligned} & \lambda x. \neg E(\llbracket \text{fake gun} \rrbracket_M^c)(x) \wedge \\ & \neg \exists e_2 [\text{MAKING}(e_2) \wedge \text{GOAL}(e_2, Q_P(\llbracket \text{gun} \rrbracket_M^c))] \wedge \\ & \exists e_3 [\text{MAKING}(e_3) \wedge \text{GOAL}(e_3, Q_P(\llbracket \text{gun} \rrbracket_M^c)(x))] \end{aligned}$$

C-structure:

$$\begin{aligned} \text{C: } & \lambda x. Q_C(\llbracket \text{gun} \rrbracket_M^c)(x) \\ \text{P: } & \lambda x. Q_P(\llbracket \text{gun} \rrbracket_M^c)(x) \\ \text{T: } & \lambda x. \neg Q_T(\llbracket \text{fake gun} \rrbracket_M^c)(x) \\ \text{A: } & \lambda x. \exists e_3 [\text{MAKING}(e_3) \wedge \text{GOAL}(e_3, Q_P(\llbracket \text{fake gun} \rrbracket_M^c)(x))] \end{aligned}$$

In other words, there both was and was not an event in which x was made to look like a gun.

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